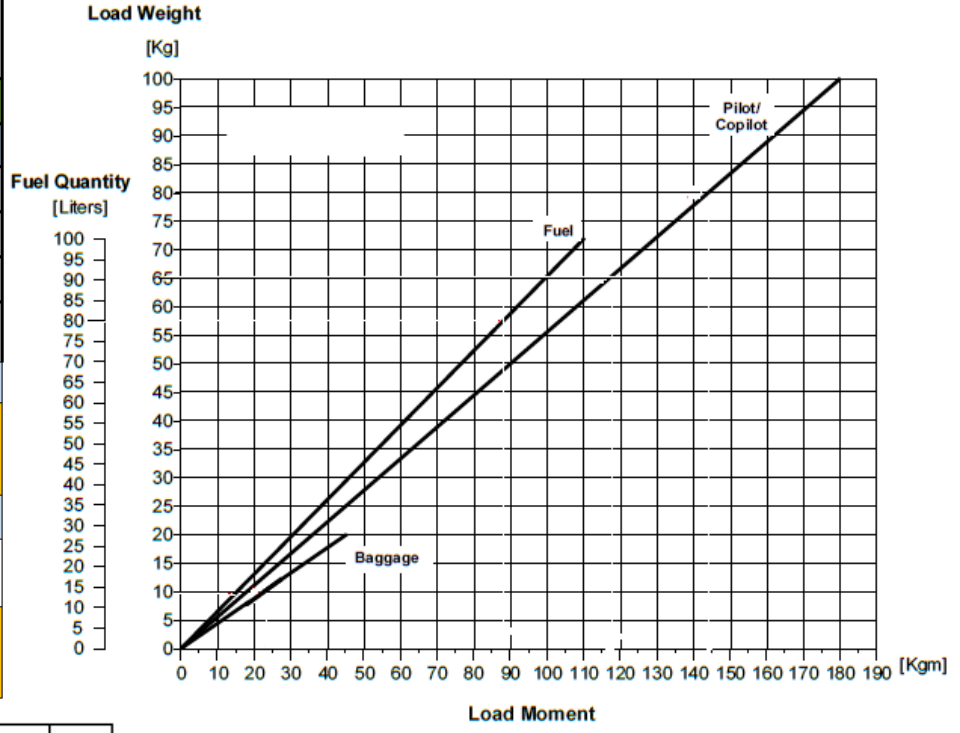
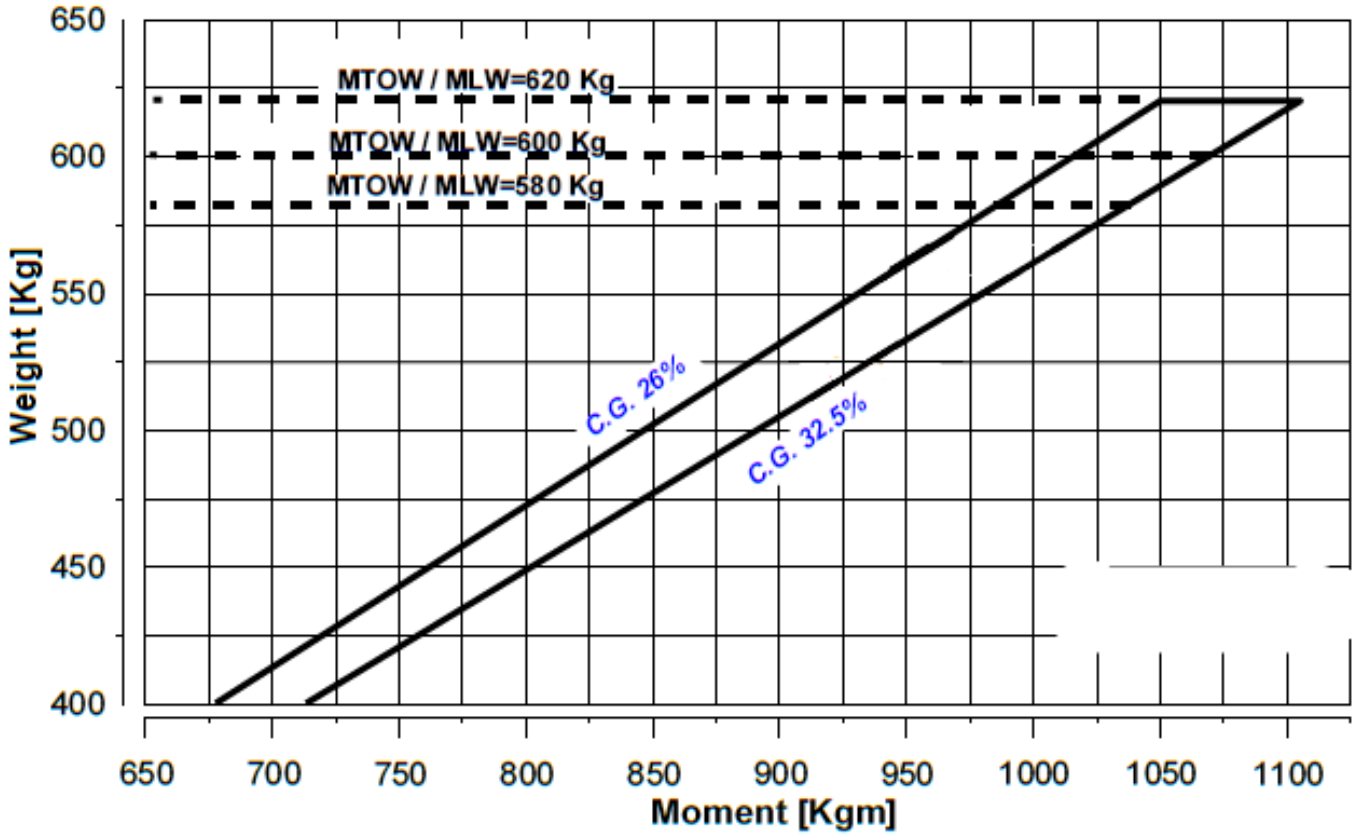
	W [kg]	Arm [m]	Moment (M) = W * Arm [kg*m]
Empty weight	**400**	1.67	**670**
LOADING			
Pilot		1.8	
Co-pilot		1.8	
Baggage		2.26	
Usable fuel Fuel (liters)*ρ_{fuel} (0.72) [kg]		1.53	
TAKE-OFF CONDITION			
Take-off condition $W_{TO} = \sum W$		$M_{TO} = \sum M$	
LANDING CONDITION			
Fuel required Fuel (liters)*ρ_{fuel} (0.72) [kg]		1.53	
Landing condition $W_L = W_{TO} - W_{fuel_req}$		$M_L = M_{TO} - M_{fuel_req}$	





Pilot/Co-pilot loading	
W [kg]	M [kg*m]
5	9
10	18
15	27
20	36
30	54
40	72
50	90
60	108
65	117
70	126
75	135
80	144
85	153
90	162
95	171
100	180

Fuel loading		
W [litres]	W [kg]	M [kg*m]
10	7.2	11.0
20	14.4	22.0
30	21.6	33.0
40	28.8	44.1
50	36.0	55.1
60	43.2	66.1
65	46.8	71.6
70	50.4	77.1
75	54.0	82.6
80	57.6	88.1
85	61.2	93.6
90	64.8	99.1
95	68.4	104.7
100	72.0	110.2

Baggage loading	
W [kg]	M [kg*m]
2	4.5
4	9.0
6	13.6
8	18.1
10	22.6
12	27.1
14	31.6
16	36.2
18	40.7
20	45.2

7. TAKE-OFF PERFORMANCES (*APPROVED DATA*)

Weight = *620 kg*

Flaps: *T/O*
Speed at Lift-Off = *42 KIAS*
Speed Over *50ft* Obstacle = *52 KIAS*
Throttle Levers: *Full Forward*
Runway: *Grass*

Corrections

Headwind: - 2.5m for each kt (*8 ft/kt*)
Tailwind: + 10m for each kt (*33ft/kt*)
Paved Runway: - *6%* to Ground Roll
Runway slope: + 5% to Ground Roll for each +1%

Pressure Altitude [ft]		Distance [m]				
		Temperature [°C]				ISA
		-25	0	25	50	
S.L.	Ground Roll	154	203	262	331	237
	At 50 ft AGL	252	335	434	553	392
1000	Ground Roll	169	223	287	364	255
	At 50 ft AGL	277	368	478	610	423
2000	Ground Roll	186	245	316	401	275
	At 50 ft AGL	305	405	526	672	456
3000	Ground Roll	204	269	348	442	296
	At 50 ft AGL	336	446	580	742	492
4000	Ground Roll	225	296	383	487	319
	At 50 ft AGL	370	492	641	820	531
5000	Ground Roll	247	327	423	538	345
	At 50 ft AGL	408	543	708	907	574
6000	Ground Roll	272	360	466	594	372
	At 50 ft AGL	450	600	783	1005	621
7000	Ground Roll	300	397	515	657	402
	At 50 ft AGL	498	664	867	1114	672
8000	Ground Roll	331	439	570	727	435
	At 50 ft AGL	551	735	962	1236	728
9000	Ground Roll	366	486	631	806	471
	At 50 ft AGL	610	815	1068	1374	790
10000	Ground Roll	405	538	700	895	510
	At 50 ft AGL	676	905	1186	1529	857

11. LANDING PERFORMANCES (*APPROVED DATA*)

Weight = *620 kg*

Flaps: *LAND*
Short Final Approach Speed = *51 KIAS*
Throttle Levers: *Idle*
Runway: *Grass*

Corrections

Headwind: - 5m for each kt (*16 ft/kt*)
Tailwind: + 11m for each kt (*36ft/kt*)
Paved Runway: - *2%* to Ground Roll
Runway slope: - 2.5% to Ground Roll for each +1%

Pressure Altitude [ft]		Distance [m]				
		Temperature [°C]				ISA
		-25	0	25	50	
S.L.	Ground Roll	141	155	170	184	164
	At 50 ft AGL	277	305	333	361	322
1000	Ground Roll	146	161	176	191	169
	At 50 ft AGL	288	317	345	374	332
2000	Ground Roll	152	167	183	198	174
	At 50 ft AGL	298	328	358	388	342
3000	Ground Roll	158	173	189	205	179
	At 50 ft AGL	309	341	372	403	352
4000	Ground Roll	164	180	196	213	185
	At 50 ft AGL	321	353	386	418	363
5000	Ground Roll	170	187	204	221	190
	At 50 ft AGL	333	367	400	434	374
6000	Ground Roll	176	194	212	230	196
	At 50 ft AGL	346	381	416	451	385
7000	Ground Roll	183	201	220	238	202
	At 50 ft AGL	359	396	432	468	397
8000	Ground Roll	190	209	228	248	209
	At 50 ft AGL	373	411	449	486	410
9000	Ground Roll	198	217	237	257	215
	At 50 ft AGL	388	427	466	505	423
10000	Ground Roll	205	226	247	267	222
	At 50 ft AGL	403	444	484	525	436